Universal Print Deployment & Print-Server Retirement — Government (GCC High)

Implementation Solution Guide

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Contents

Executive Summary — Universal Print Deployment & Print-Server Retirement (Government / GCC High) 5

- The opportunity . . . 5
- Why a Government (GCC High) edition is different . . . 5
- Two availability facts leadership should know . . . 5
- Approach . . . 5
- Value . . . 5
- Bottom line . . . 6

Solution Overview — Universal Print Deployment & Print-Server Retirement (Government / GCC High) 6

- Availability up front — read this first . . . 6
- What this solution does . . . 6
- The deployment model at a glance . . . 7
- Commercial vs GCC High — what's different . . . 7
- Who this is for . . . 8
- What you get . . . 8
- The bottom line . . . 9

Requirements Guide — Universal Print on GCC High 9

- R1. A GCC High tenant (not Commercial, not DoD) . . . 9
- R2. Universal Print licensing — for every user AND every admin . . . 9
- R3. Administrative roles (GCC High) . . . 10
- R4. Universal Print service is enabled for the tenant . . . 10
- R5. Connector host (for existing / non-cloud-ready printers) . . . 11
- R6. Network access to Universal Print GCC High endpoints . . . 11
- R7. Entra-joined / Intune-managed Windows endpoints . . . 11
- R8. Windows client version (GCC High native vs registry-keyed) . . . 12
- R9. Microsoft Graph PowerShell (US Gov) . . . 12
- R10. Current print environment inventory (the "from" state) . . . 12
- Requirements checklist (all must pass before Configure) . . . 13

Implementation Guide — Universal Print on GCC High 13

- Naming convention (create once, reuse everywhere) . . . 14

PHASE 1 — ASSESS 14

- Step 1 — Confirm the cloud and licensing . . . 14
- Step 2 — Baseline the current print environment (read-only) . . . 14

PHASE 2 — CONFIGURE 15

Step 3 — Open the Universal Print portal (GCC High) . . . 15
Step 4 — Prepare the connector host . . . 15
Step 5 — Register the connector . . . 15
Step 6 — Register and share the first printers . . . 16
Step 7 — Create targeting groups . . . 16
PHASE 3 — PILOT 16
Step 8 — Prepare pilot clients . . . 16
Step 9 — Deploy the printer to pilot users . . . 17
Step 10 — First print jobs . . . 17
PHASE 4 — VALIDATE 17
Step 11 — Prove the pilot . . . 17
PHASE 5 — EXPAND & RETIRE THE PRINT SERVER 18
Step 12 — Roll out in waves . . . 18
Step 13 — Retire the legacy print server(s) . . . 18
Endpoint quick-reference (GCC High) . . . 18
Testing & Validation — Universal Print on GCC High 19
Test set A — Assess gate . . . 19
Test set B — Configure gate . . . 19
Test set C — Pilot gate . . . 20
Test set D — Validate gate . . . 20
Test set E — Expand & retirement gate (per wave) . . . 21
Regression / ongoing checks . . . 21
Troubleshooting — Universal Print on GCC High 22
T1. Sign-in fails / "account doesn't exist" / redirect error when connecting Graph . . . 22
T2. Universal Print blade shows "not available" or licensing banner in the portal . . . 22
T3. Printer discovery fails on Windows 10 / older Windows 11 clients (works on Win 11 22H2+) . . . 22
T4. Connector won't register or drops offline . . . 23
T5. Endpoints unreachable / intermittent print failures . . . 23
T6. Printer registered but users can't see it . . . 23
T7. Delegated (scoped) admin not working in GCC High . . . 24
T8. macOS users can't print . . . 24
T9. Print jobs succeed but don't appear in reporting / job pool concerns . . . 24
T10. "It worked in the lab (Commercial) but not in production (GCC High)" . . . 24
T11. DoD tenant — nothing works . . . 24
Escalation checklist (before opening a Microsoft support case) . . . 25
Best Practices — Universal Print on GCC High 25

1. Stay in the sovereign cloud — every time . . . 25
2. License discipline (Government-specific) . . . 25
3. Pilot-first, always gate . . . 26
4. Retire print servers safely . . . 26
5. Respect GCC High parity gaps (design around them) . . . 26
6. Connector hygiene . . . 26
7. Client-version strategy . . . 26
8. Naming, targeting, and least privilege . . . 27
9. Change safety . . . 27
10. Monitoring and lifecycle . . . 27
Deliverables Summary — Universal Print Deployment & Print-Server Retirement (GCC High) 27
Documents . . . 27
PowerShell scripts (all connect to US Gov Graph, `-Environment USGov`) . . . 28
How the pieces fit the phases . . . 29
Outcome . . . 29
Appendix A — PowerShell Scripts 30

Executive Summary — Universal Print Deployment & Print-Server Retirement (Government / GCC High)

The opportunity

On-premises Windows print servers are a recurring cost and risk: OS patching, per-site hardware, driver management, and single points of failure. **Universal Print** — Microsoft's cloud print service — removes that infrastructure by publishing printers from Microsoft Entra ID and delivering print jobs through the cloud to managed Windows devices. This solution deploys it on a **GCC High** tenant and retires the legacy print servers, safely.

Why a Government (GCC High) edition is different

Universal Print in the US sovereign cloud is **not** the Commercial service with a different sign-in. It runs on dedicated GCC High infrastructure with **different portal URLs, different service endpoints, different Graph endpoints, and different licensing** — and it has a few feature-parity gaps versus Commercial. Configuring a GCC High tenant with Commercial steps causes **silent failures**. This edition corrects every portal (`*.us`), every endpoint (`*.print.azure.us`), and every script (`-Environment USGov`).

Two availability facts leadership should know

- **Universal Print is generally available in GCC and GCC High** — this deployment is fully supported for a GCC High tenant.
- **Universal Print is not available in DoD** at this time. Tenants in the DoD cloud cannot yet deploy it. (Source: Microsoft. *Verify against the Microsoft 365 roadmap.*)
- **Some features lag in GCC High** — notably **Delegated Administration** and **macOS printing** — so the deployment is designed to work around those gaps rather than assume them.

Approach

The deployment follows a disciplined, pilot-first model with a **gate at every phase**:

Assess → **Configure** → **Pilot** → **Validate** → **Expand.**

No print server is retired until every printer it hosted is proven working in Universal Print. Five bundled PowerShell scripts (all US-Gov-Graph aware) automate discovery, validation, reporting, and safe (`-WhatIf`-first) remediation so the work is repeatable and auditable — and executable by a novice-to-intermediate administrator.

Value

- **Eliminates print-server infrastructure** — patching, hardware refresh, and per-site management cost.
- **Correct the first time on GCC High** — no rework from Commercial steps that silently fail.

- **Low-risk rollout** — pilot and wave gates with rollback; servers retired only after proof.
- **Compliance-aligned** — Universal Print is FedRAMP-certified via Office 365 and runs on the US sovereign cloud with TLS 1.2 encryption (verify accreditation scope for your tenant).
- **A fraction of consulting cost** — self-guided, tested, and reusable.

Bottom line

Universal Print deployed correctly on GCC High — right portals, right endpoints, parity gaps handled — and proven on a pilot before you scale, so legacy print servers are retired without an outage.

Solution Overview — Universal Print Deployment & Print-Server Retirement (Government / GCC High)

Availability up front — read this first

Universal Print IS available in GCC High. Microsoft announced **general availability of Universal Print in GCC and GCC High**, and the service is documented as running on the US sovereign cloud with its own `*.print.azure.us` service endpoints and US Gov identity (`login.microsoftonline.us`). This solution is written for a **GCC High** tenant.

Two availability facts you must internalize before you begin — both are current Microsoft statements (see `gcc-high-parity-notes.md` for citations):

- **DoD is NOT supported.** Universal Print is *not* available in the Microsoft 365 **DoD** cloud at time of authoring. If your tenant is DoD (not GCC High), **stop** — this solution does not apply, and there is no supported Universal Print deployment for you yet. *Verify for GCC High/DoD before scoping.*
- **GCC High has feature-parity gaps vs Commercial.** Microsoft states it is "aiming for feature parity," but at authoring time specific features lag in GCC High — notably **Delegated Administration** ("coming to GCC High soon") and **macOS client printing** ("not yet available for GCC High"). These are called out inline where they matter and consolidated in `gcc-high-parity-notes.md`. Do not assume a Commercial how-to step works unchanged.

If you were handed the Commercial edition of this solution by mistake, it will point you at `portal.azure.com`, `graph.microsoft.com`, and `login.microsoftonline.com` — **none of which serve a GCC High tenant.** This edition corrects every portal, endpoint, and Graph environment for the US sovereign cloud.

What this solution does

Universal Print is Microsoft's cloud-based print service. It removes the need for on-premises **Windows print servers**, print-server OS patching, and per-site print infrastructure by publishing printers from Microsoft Entra ID and delivering print jobs through the cloud to Entra-joined Windows endpoints. This solution takes a GCC High tenant from "we still run print servers" to "printers are published in Universal Print, endpoints print through the cloud, and the legacy print server is retired" — safely, on a pilot-first path, with a rollback at every gate.

The deployment follows the standard repeatable model used across all Infrastructure solutions:

Assess → Configure → Pilot → Validate → Expand

Each phase has an explicit **gate** — you do not proceed until the prior phase passes its validation criteria. The five bundled PowerShell scripts (all connecting to **US Gov Graph** with `-Environment USGov`) automate discovery, validation, reporting, and safe remediation so a novice administrator can execute the whole sequence and prove each gate.

The deployment model at a glance

Phase	Goal	Gate to pass before moving on
Assess	Inventory current print servers, printers, drivers, and per-printer job volume; confirm GCC High licensing and endpoints.	`UPRTG-Discovery.ps1` runs clean; every prerequisite in `02-requirements.md` verified.
Configure	Enable Universal Print, register the connector, publish and share the first printers, set targeting groups.	Printers appear as **Shared** in the Universal Print portal (`portal.azure.us`).
Pilot	Add one pilot printer + the pilot security group; users print end-to-end through the cloud.	Pilot users complete real print jobs; `UPRTG-Validation.ps1` passes for the pilot scope.
Validate	Confirm print success rate, driver behavior, secure release, and reporting for the pilot.	No open Sev-1/Sev-2 issues; sign-off recorded.
Expand	Roll out remaining printers/sites in waves; retire the legacy print server(s).	Each wave validated before the next; print server decommissioned only after its printers are proven in Universal Print.

Commercial vs GCC High — what's different

This is the section that keeps a government tenant from being configured with Commercial steps that silently fail. Everything below is specific to the **US sovereign cloud**.

Area	Commercial	GCC High (this edition)
Azure / Universal Print admin portal	`portal.azure.com`	`portal.azure.us` (Azure Government)
Microsoft 365 admin center	`admin.microsoft.com`	`portal.office365.us`
Microsoft Intune / Endpoint admin	`intune.microsoft.com`	`intune.microsoft.us` (endpoint.microsoft.us)
Microsoft Entra admin	`entra.microsoft.com`	`entra.microsoft.us` (or Entra blade in `portal.azure.us`)
Sign-in / identity authority	`login.microsoftonline.com`	`login.microsoftonline.us`

Area	Commercial	GCC High (this edition)
Microsoft Graph endpoint	`graph.microsoft.com`	`graph.microsoft.us`
Graph PowerShell environment	`-Environment Global` (default)	`-Environment USGov`
Universal Print service endpoints	`*.print.microsoft.com`	`*.print.azure.us` (print, register, discovery, notification, graph)
Licensing SKUs	M365 E3/E5, Windows E3/E5	**G3/G5 Gov** — `SPE_E3_USGOV_GCCHIGH`, `SPE_E5_USGOV_GCCHIGH`, or `Universal_Print_for_GCCHIGH` standalone
Windows client support	All supported Windows	**Windows 11 22H2+ native**; Windows 10 / older Win 11 require three registry keys to point at the US Gov authority/Graph (see `03-implementation.md`)
Delegated Administration	Available	**Coming to GCC High "soon"** — *verify for GCC High* before designing scoped admin
macOS client printing	Available	**Not yet available for GCC High** — *verify for GCC High*
DoD availability	N/A	**Universal Print not available in DoD at all**

Three practical consequences of that table:

1. **Every portal URL in this edition ends in `.us`.** If a screen you land on ends in `.com`,

you are in the wrong cloud and your GCC High credentials will not work there.

1. **Every script connects with `-Environment USGov`.** A script that omits this authenticates

against the Commercial identity endpoint and fails for a GCC High account.

1. **Licensing is mandatory for administrators too.** In Government, **all users — including

the admins who configure Universal Print — must hold an eligible Universal Print license** (G3/G5 Gov or the standalone SKU). This is a stricter rule than many admins expect.

Who this is for

A novice-to-intermediate administrator on a **GCC High** Microsoft 365 tenant who needs to stand up Universal Print correctly the first time and retire aging print servers without an outage — and without being tripped up by Commercial documentation that points at the wrong cloud.

What you get

- **This overview** and a full **Requirements Guide** (`02-requirements.md`) with verify +

implement steps for every prerequisite on GCC High.

- A step-by-step **Implementation Guide** (`03-implementation.md`) using the `.us` portals,

with GCC-High-only caveats placed at the exact step where they bite.

- **Five PowerShell scripts** — Discovery, Validation, Report, Remediation, and

Set-ScopeTargeting — all US-Gov-Graph aware.

- A **Testing guide** (`05-testing.md`), a **Troubleshooting guide** (`06-troubleshooting.md`),

Best practices (`07-best-practices.md`), a **Deliverables summary** (`08-deliverables-summary.md`), an **Executive summary**, and the cited `gcc-high-parity-notes.md` parity reference.

The bottom line

Universal Print deployed correctly on GCC High — right portals, right endpoints, parity gaps handled — and proven on a pilot before you scale, so the legacy print server is retired only after its printers work in the cloud.

Requirements Guide — Universal Print on GCC High

Confirm **every** requirement below before you configure anything. The category-level prerequisites live in `global-requirements-infrastructure.md`; this guide adds the **GCC-High-specific** requirements and, where the global list assumes Commercial URLs, **overrides them with the US sovereign-cloud equivalents**. Each entry gives you: what it is, why it matters, how to **verify** it, and how to **implement** it if it is missing.

> ***Portal reminder.*** *On GCC High, every admin portal ends in* `.us`*. If a page you open ends in* `.com`*, you are in the wrong cloud — sign out and use the* `.us` *URL listed here.*

R1. A GCC High tenant (not Commercial, not DoD)

- **What:** Your Microsoft 365 tenant must be a **GCC High** tenant.

- **Why:** Universal Print is GA in **GCC and GCC High** but **not in DoD**. Commercial steps

and endpoints do not serve GCC High. This whole solution assumes GCC High.

- **Verify:** Sign in at `https://portal.office365.us` (Microsoft 365 Government admin

center). Confirm the tenant loads there and **not** at `admin.microsoft.com`. In `https://portal.azure.us` confirm the Azure Government portal loads with your account.

- **Implement:** If you are on Commercial, this GCC High edition does not apply — use the

Commercial edition. If you are on **DoD**, **stop**: Universal Print is not available in DoD; there is no supported deployment. *Verify for GCC High/DoD.*

R2. Universal Print licensing — for every user AND every admin

• **What:** An eligible Universal Print license assigned to **all** users who will print **and

to every administrator who will configure the service**.

• **Why:** In Government, Microsoft explicitly requires that **all users, including

administrators, be assigned an eligible license** to use *or configure* Universal Print. Each license also contributes print jobs to a shared monthly pool.

• **Verify:** In `https://portal.office365.us` → **Billing** → **Licenses**, confirm one of the

GCC High SKUs below is owned and assigned. In **Users** → **Active users**, spot-check that the admin account and pilot users each have it.

• **Implement:** Assign one of these GCC High SKUs (part numbers as published by Microsoft):

• **Microsoft 365 E3 GCC High** — `SPE_E3_USGOV_GCCHIGH` (100 jobs/user/month)

• **Microsoft 365 E5 GCC High** — `SPE_E5_USGOV_GCCHIGH` (100 jobs/user/month)

• **Universal Print for GCC High (standalone)** — `Universal_Print_for_GCCHIGH_USGOV_GCCHIGH` (5 jobs/month)

After assignment, **allow up to 24 hours** for Universal Print access to appear.

R3. Administrative roles (GCC High)

• **What:** The operator needs a role that can enable Universal Print and manage printers:

Printer Administrator (least privilege for day-to-day) — plus **Global Administrator** for the one-time enable and connector registration if required.

• **Why:** Publishing printers, sharing them, and registering the connector require elevated

rights. Universal Print exposes **Printer Administrator** and **Printer Technician** roles for least-privilege operation.

• **Verify:** In `https://entra.microsoft.us` (or the Entra blade inside `portal.azure.us`)

→ **Roles and administrators**, confirm your account holds **Printer Administrator** (and, for first-time enablement, **Global Administrator**).

• **Implement:** Assign **Printer Administrator** to the operator. Reserve **Global

Administrator **for the tenant-wide enable step only.** Note: ***Delegated Administration* (scoped Printer Administrator via Entra administrative units) is** coming to GCC High "soon"** — do not design a scoped-admin model that depends on it yet; *verify for GCC High.*

R4. Universal Print service is enabled for the tenant

• **What:** Universal Print must be turned on for the tenant.

• **Why:** Nothing can be published until the service is enabled.

• **Verify:** In `https://portal.azure.us`, search **Universal Print** and open the blade.

It should load without a "service not available / license required" banner.

• **Implement:** With an eligible license (R2) assigned, open the **Universal Print** blade in

`portal.azure.us`; enablement is automatic once licensing propagates (allow 24 h).

R5. Connector host (for existing / non-cloud-ready printers)

- **What:** A Windows host running the **Universal Print connector** to bridge existing

network/USB printers into the cloud. Recommended: **Windows 11 64-bit Pro/Enterprise (build 22631+)** or **Windows Server 2025**; supported (not recommended): Windows 10 (19045+) / Windows Server 2022. Requires **.NET Framework 4.8+**.

- **Why:** Printers that are not "Universal Print ready" reach the cloud only through the

connector. Microsoft notes **no Universal-Print-ready printers connect directly in GCC/GCC High yet** — so in GCC High you will almost always use the connector.

- **Verify:** Confirm the host is 64-bit, on a recommended build, has .NET 4.8+

(`Get-ItemProperty 'HKLM:\SOFTWARE\Microsoft\NET Framework Setup\NDP\v4\Full' -Name Release`), runs 24×7 (sleep/hibernate disabled), and has constant internet.

- **Implement:** Provision a dedicated always-on host; disable power saving; install .NET 4.8+;

download the connector from `https://aka.ms/UPConnector`. **Do not clone** a registered connector host (clones interfere and block printing).

R6. Network access to Universal Print GCC High endpoints

- **What:** Outbound HTTPS (TLS 1.2) from the connector host and client endpoints to the **GCC

High** Universal Print endpoints and US Gov identity/Graph.

- **Why:** GCC High uses **dedicated** `*.print.azure.us` **endpoints** and the **US Gov identity

authority**, not the Commercial `*.print.microsoft.com` / `login.microsoftonline.com` URLs. If your proxy/firewall only allows Commercial URLs, GCC High printing fails silently.

- **Verify:** From the connector host and a client, confirm outbound 443 to:
- `https://print.print.azure.us`
- `https://register.print.azure.us`
- `https://discovery.print.azure.us`
- `https://notification.print.azure.us`
- `https://graph.print.azure.us`
- `https://login.microsoftonline.us/`
- `https://graph.microsoft.us`
- **Implement:** Allow the URLs above through the proxy/firewall. If the connector uses a proxy,

set the proxy for the **LocalSystem** account with `bitsadmin` (per Microsoft's connector guidance) — a user-context proxy is not enough for the connector service.

R7. Entra-joined / Intune-managed Windows endpoints

• **What:** The Windows devices that will print must be **Microsoft Entra joined** (or hybrid) and, ideally, **Intune-managed** (via `https://intune.microsoft.us`).

• **Why:** Universal Print discovers and deploys printers to Entra-joined devices; Intune is how you deploy printers and the Win10 registry keys (R8) at scale.

• **Verify:** In `https://intune.microsoft.us` → **Devices** → **All devices**, confirm the device appears and is compliant. On the device, `dsregcmd /status` shows `AzureAdJoined : YES`.

• **Implement:** Enable automatic enrollment and Entra join per your standard build. See the global prerequisites for baseline enrollment steps.

R8. Windows client version (GCC High native vs registry-keyed)

• **What:** Client Windows version determines whether Universal Print works natively.

• **Why:** In GCC High, **Windows 11 22H2 and later is supported natively.** **Windows 10 and earlier Windows 11 **require** three registry keys** that point the print stack at the US Gov authority and Graph — without them, those clients try Commercial endpoints and fail.

• **Verify:** `winver` on each client. For any client below Win 11 22H2, confirm the three `HKCU\SOFTWARE\Microsoft\Print\MPS` values are present (see `03-implementation.md`, Step 8).

• **Implement:** Prefer upgrading pilot devices to **Windows 11 22H2+**. Where you must support Windows 10 / older Win 11, deploy the three registry keys (OAuthAuthority = `https://login.microsoftonline.us/common`, MSGraphBaseUrl = `https://graph.microsoft.us/v1.0`, MSGraphResourceId = `https://graph.microsoft.us`) via Intune to the user context.

R9. Microsoft Graph PowerShell (US Gov)

• **What:** The **Microsoft.Graph** PowerShell module, used by the five bundled scripts, able to connect with `-Environment USGov`.

• **Why:** The scripts read printer/share/connector state from `graph.microsoft.us`. Without `-Environment USGov` the module authenticates against Commercial and your GCC High account fails.

• **Verify:** `Get-Module Microsoft.Graph -ListAvailable` shows the module; then `Connect-MgGraph -Environment USGov -Scopes "Printer.Read.All"` completes sign-in against `login.microsoftonline.us`.

• **Implement:** `Install-Module Microsoft.Graph -Scope CurrentUser`. All bundled scripts pass `-Environment USGov` for you.

R10. Current print environment inventory (the "from" state)

• **What:** A documented inventory of existing **print servers, shared printers, drivers, ports,

and per-printer job volume**.

• **Why:** You retire print servers only after their printers are proven in Universal Print.

You cannot plan waves or size the job pool without this baseline.

• **Verify:** Run `UPRTG-Discovery.ps1` (read-only) to enumerate what Universal Print already sees, and cross-reference against your print-server `Get-Printer` exports.

• **Implement:** On each print server, export `Get-Printer`, `Get-PrinterDriver`, and `Get-PrinterPort`; record monthly job counts per printer to size the shared job pool.

Requirements checklist (all must pass before Configure)

#	Requirement	Verified at (GCC High URL / command)
R1	GCC High tenant (not DoD)	`portal.office365.us`, `portal.azure.us`
R2	UP license for users **and** admins	`portal.office365.us` → Licenses
R3	Printer Administrator role	`entra.microsoft.us` → Roles
R4	Universal Print enabled	`portal.azure.us` → Universal Print
R5	Connector host ready	Win 11 22631+/WS2025, .NET 4.8+
R6	`*.print.azure.us` endpoints allowed	outbound 443 test
R7	Entra-joined / Intune-managed clients	`intune.microsoft.us` → Devices
R8	Win 11 22H2+ (or registry keys)	`winver` / registry check
R9	Graph PowerShell `-Environment USGov`	`Connect-MgGraph -Environment USGov`
R10	Print environment inventory captured	`UPRTG-Discovery.ps1`

When every row is verified, proceed to `03-implementation.md`.

Implementation Guide — Universal Print on GCC High

This guide takes you from an enabled GCC High tenant to published, shared, pilot-validated printers and, finally, a retired print server — following **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** with a **gate** at each phase. Every portal URL ends in `.us`. Every script connects with `-Environment USGov`. Object names below are **reused consistently** — create them once and reference them at every later step.

> ***Before you start:*** *complete* `02-requirements.md`*. Do not begin Configure until all ten requirements verify. GCC-High-only caveats are flagged* ***[GCC High]*** *at the step where they bite.*

Naming convention (create once, reuse everywhere)

Object	Name to use	Where
Pilot security group	`UP-Pilot-Users`	Entra (`entra.microsoft.us`)
Expand wave group (per wave)	`UP-Wave1-Users`, `UP-Wave2-Users`, …	Entra
Connector	`UP-Connector-Site1`	Connector host / UP portal
Pilot printer share	`UP-Pilot-Printer-01`	Universal Print portal
Production printer shares	`UP-<Site>-<Model>-NN`	Universal Print portal

PHASE 1 — ASSESS

Step 1 — Confirm the cloud and licensing

1. Sign in to `https://portal.office365.us` (Microsoft 365 Government admin center).
2. **Billing** → **Licenses** — confirm a GCC High UP SKU is present (`SPE_E3_USGOV_GCCHIGH`, `SPE_E5_USGOV_GCCHIGH`, or `Universal_Print_for_GCCHIGH_USGOV_GCCHIGH`).
1. **Users** → **Active users** — confirm **your admin account** and each intended pilot user has the license assigned. **[GCC High]** Admins without a license cannot even configure the service.
1. If you just assigned licenses, **wait up to 24 hours** before expecting Universal Print to appear.

Expected result: Eligible license owned and assigned to admin + pilot users.

Step 2 — Baseline the current print environment (read-only)

1. On the admin workstation, install Graph PowerShell if needed:

`Install-Module Microsoft.Graph -Scope CurrentUser`.

1. Run the discovery script (read-only, US Gov Graph):

```
powershell -ExecutionPolicy Bypass -File .\UPRTG-Discovery.ps1
```

Sign in when prompted — the browser opens `login.microsoftonline.us`. The script reports whether Universal Print is reachable, lists any already-registered printers/shares/connectors, and checks the GCC High endpoints.

1. On each legacy print server, export the "from" state:

```
Get-Printer | Export-Csv .\printers.csv -NoTypeInformation
Get-PrinterDriver | Export-Csv .\drivers.csv -NoTypeInformation
Get-PrinterPort | Export-Csv .\ports.csv -NoTypeInformation
```

1. Record **monthly job volume per printer** to size the shared job pool (each G3/G5 Gov license contributes 100 jobs/month to the pool).

Gate (Assess → Configure): `UPRTG-Discovery.ps1` runs clean against `graph.microsoft.us`, and every `02-requirements.md` row is verified.

PHASE 2 — CONFIGURE

Step 3 — Open the Universal Print portal (GCC High)

1. Sign in to `https://portal.azure.us` (Azure **Government** portal — note the `.us`).
2. Search **Universal Print** and open the blade. It must load without a licensing/availability banner. **[GCC High]** If it shows "not available," you are either unlicensed (Step 1) or in the wrong cloud (you opened `portal.azure.com`).

Expected result: Universal Print blade loads in `portal.azure.us`.

Step 4 — Prepare the connector host

[GCC High] No Universal-Print-ready printers connect directly in GCC High yet, so you will use the **connector** for existing printers.

1. On the always-on host (`UP-Connector-Site1`), confirm **.NET Framework 4.8+** and that sleep/hibernate are disabled.
1. **[GCC High]** Ensure outbound 443 to the `*.print.azure.us` endpoints, `login.microsoftonline.us`, and `graph.microsoft.us` (see `02-requirements.md` R6). If a proxy is used, set it for the **LocalSystem** account with `bitsadmin` — the connector runs as a service.
1. Download the connector from `https://aka.ms/UPConnector` and install it.

Expected result: Connector app installed on `UP-Connector-Site1`.

Step 5 — Register the connector

1. Launch the connector app; click **Login** and sign in with your GCC High admin account (you are redirected to `login.microsoftonline.us`).
1. In **Connector name**, enter `UP-Connector-Site1` and click **Register**.
2. After registration, the app lists local/network printers available to register.
3. Configure connector options: leave **Automatically collect diagnostics** per your policy, and enable **Hybrid AD configuration** only if on-prem AD identity must be used for print jobs (and only after meeting its prerequisites).

[GCC High] Warning: never **clone** a registered connector host — parallel clones interfere and block printing.

Expected result: `UP-Connector-Site1` shows **Registered** and lists printers.

Step 6 — Register and share the first printers

1. In the connector app, select the printer(s) to onboard and **Register** them with Universal Print.
1. In `https://portal.azure.us` → **Universal Print** → **Printers**, confirm the registered printer appears.
1. Select the printer → **Share**. Give the share a reusable name, e.g. `UP-Pilot-Printer-01`.
2. On the **Share** blade, choose **who** can access it — for the pilot, you will grant the `UP-Pilot-Users` group in Step 7.

Expected result: The printer shows as **Shared** in Universal Print with share name `UP-Pilot-Printer-01`.

Step 7 — Create targeting groups

1. In `https://entra.microsoft.us` → **Groups** → **New group** → **Security**, create `UP-Pilot-Users` and add the pilot users.
1. (Optional, recommended) Pre-create empty `UP-Wave1-Users`, `UP-Wave2-Users` groups for later expansion.
1. You can build/populate these groups repeatably with the bundled script:

```
powershell -ExecutionPolicy Bypass -File .\UPRTG-Set-ScopeTargeting.ps1 -GroupName "UP-Pilot-Users" -Wh
```

Review the `-WhatIf` plan, then re-run **without** `-WhatIf` to apply.

1. On the `UP-Pilot-Printer-01` share, grant access to `UP-Pilot-Users`.

Gate (Configure → Pilot): printer is **Shared**, `UP-Pilot-Users` exists and is granted on the share.

PHASE 3 — PILOT

Step 8 — Prepare pilot clients

1. Confirm each pilot device is **Entra-joined** and, ideally, **Intune-managed** (`intune.microsoft.us`).
1. **[GCC High]** Check `winver`:

• **Windows 11 22H2 or later** → natively supported, no registry keys needed.

• **Windows 10 / earlier Windows 11** → deploy these **three registry keys** to the **user**

context (via Intune or a user-context script), then have the user sign out/in:

```
reg add "HKCU\SOFTWARE\Microsoft\Print\MPS" /t REG_SZ /v "OAuthAuthority"  /d "https://login.microsof
reg add "HKCU\SOFTWARE\Microsoft\Print\MPS" /t REG_SZ /v "MSGraphBaseUrl"  /d "https://graph.microsof
reg add "HKCU\SOFTWARE\Microsoft\Print\MPS" /t REG_SZ /v "MSGraphResourceId" /d "https://graph.micros
```

Without these, older clients try Commercial endpoints and printer discovery fails.

Expected result: Pilot clients are Entra-joined and (if pre-22H2) carry the three US Gov registry keys.

Step 9 — Deploy the printer to pilot users

Two supported methods:

- **User self-service (quickest for pilot):** on a pilot client, **Settings → Bluetooth &

devices → Printers & scanners → Add device → Add a Universal Print printer**. The user (signed in with their GCC High account) sees `UP-Pilot-Printer-01` because they are in `UP-Pilot-Users`.

- **Intune-managed deployment (repeatable):** In `https://intune.microsoft.us` use the

Universal Print printer provisioning (Printers) capability to assign `UP-Pilot-Printer-01` to the `UP-Pilot-Users` group so the printer is installed automatically.

Expected result: Pilot users see and can select `UP-Pilot-Printer-01`.

Step 10 — First print jobs

1. Have pilot users print real documents (a mix: mono, color, duplex).
2. Confirm jobs complete and appear in **`portal.azure.us` → Universal Print → Usage and

Reports**.

1. Run validation for the pilot scope:

```
powershell -ExecutionPolicy Bypass -File .\UPRTG-Validation.ps1
```

Gate (Pilot → Validate): pilot users complete real jobs; `UPRTG-Validation.ps1` passes.

PHASE 4 — VALIDATE

Step 11 — Prove the pilot

1. Run the reporting script to capture evidence:

```
powershell -ExecutionPolicy Bypass -File .\UPRTG-Report.ps1
```

It writes an HTML/CSV report of printers, shares, connector state, and pilot targeting.

1. Confirm: print success rate acceptable, drivers behave, secure release works if used, and

Usage and Reports shows the pilot jobs.

1. Address any issues with `06-troubleshooting.md`. For safe fixes, use:

```
powershell -ExecutionPolicy Bypass -File .\UPRTG-Remediation.ps1 -WhatIf
```

Review the `-WhatIf` plan before re-running without it.

1. Record **sign-off**.

Gate (Validate → Expand): no open Sev-1/Sev-2 issues; pilot sign-off recorded.

PHASE 5 — EXPAND & RETIRE THE PRINT SERVER

Step 12 — Roll out in waves

1. For each wave, create/populate `UP-Wave1-Users` (then Wave2, …) with `UPRTG-Set-ScopeTargeting.ps1` (run `-WhatIf` first).
1. Register and **share** the wave's printers (`UP-<Site>-<Model>-NN`), grant the wave group, and deploy via Intune.
1. **Validate each wave** (`UPRTG-Validation.ps1` + `UPRTG-Report.ps1`) before starting the next.

Step 13 — Retire the legacy print server(s)

Retire a print server only after every one of its printers is proven in Universal Print.

1. Confirm each printer that lived on the server is now Shared in Universal Print and validated in a wave.
1. Redirect/print-test from representative clients to confirm no dependency remains on the old server (GPO print deployment, mapped `\\server\printer` shares).
1. Remove the server's printer shares from GPO/scripts; monitor for a defined soak period (e.g. 1–2 weeks) with the server still powered but shares removed.
1. When no fallback is needed, decommission the print-server role / host.

[GCC High] parity check before retirement: if your retirement design assumed **Delegated Administration** (scoped admin) or **macOS** printing, re-confirm current GCC High availability first — both were **lagging in GCC High** at authoring (*verify for GCC High*). Do not retire a server that is the only path for a not-yet-available GCC High capability your users depend on.

Final result: printers published in Universal Print on GCC High, endpoints printing through the cloud, legacy print server(s) retired — each step proven before the next.

Endpoint quick-reference (GCC High)

Service	GCC High endpoint
Print	`https://print.print.azure.us`
Registration	`https://register.print.azure.us`
Discovery	`https://discovery.print.azure.us`
Notification	`https://notification.print.azure.us`

Service	GCC High endpoint
Graph (print)	`https://graph.print.azure.us`
Identity	`https://login.microsoftonline.us/`
Microsoft Graph	`https://graph.microsoft.us`
Graph PowerShell	`Connect-MgGraph -Environment USGov`

Testing & Validation — Universal Print on GCC High

Testing follows the same **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** gates as the implementation. Each phase has explicit **pass/fail** criteria. Do not advance a phase until its tests pass. All Graph-based checks run against `graph.microsoft.us` via `-Environment USGov`; all portal checks use `.us` URLs.

> *Run the bundled scripts in this order at each gate: **Discovery** → **Validation** → **Report**. `UPRTG-Discovery.ps1`, `UPRTG-Validation.ps1`, and `UPRTG-Report.ps1` are all **read-only**.*

Test set A — Assess gate

#	Test	How	Pass	Fail action
A1	Correct cloud	Open `portal.office365.us` and `portal.azure.us`; both load with your account	Both `.us` portals load	If only `.com` works, you are on Commercial — wrong edition
A2	Licensing (users + admin)	`portal.office365.us` → Licenses / Users	GCC High UP SKU assigned to admin **and** pilot users	Assign SKU; wait ≤24 h (R2)
A3	UP reachable via US Gov Graph	`UPRTG-Discovery.ps1`	Script signs in at `login.microsoftonline.us`, reports UP reachable	See `06-troubleshooting.md` T1/T2
A4	Endpoint reachability	Discovery script endpoint check + manual 443 test to `*.print.azure.us`	All GCC High endpoints reachable	Open firewall/proxy (R6)

Gate: A1–A4 all pass.

Test set B — Configure gate

#	Test	How	Pass	Fail action
B1	Connector registered	`portal.azure.us` → Universal Print → Connectors	`UP-Connector-Site1` shows **Registered**	Re-register (Impl. Step 5); check LocalSystem proxy
B2	Printer registered	UP → Printers	Printer listed	Re-register from connector app
B3	Printer shared	UP → Printer → Share	Share `UP-Pilot-Printer-01` shows **Shared**	Re-share (Impl. Step 6)
B4	Targeting group exists	`entra.microsoft.us` → Groups	`UP-Pilot-Users` exists, has members, granted on share	`UPRTG-Set-ScopeTargeting.ps1 -WhatIf` then apply

Gate: B1–B4 all pass.

Test set C — Pilot gate

#	Test	How	Pass	Fail action
C1	Client readiness	`winver` on each pilot device	Win 11 22H2+ **or** three US Gov registry keys present (pre-22H2)	Deploy registry keys (Impl. Step 8)
C2	Printer visible to user	Pilot user: Add a Universal Print printer	`UP-Pilot-Printer-01` appears	Confirm group membership + share grant
C3	End-to-end print	Print mono, color, duplex	All jobs complete at the device	See `06-troubleshooting.md`
C4	Job accounting	`portal.azure.us` → UP → Usage and Reports	Pilot jobs appear	Wait for report refresh; recount
C5	Automated validation	`UPRTG-Validation.ps1`	Script reports PASS for pilot scope	Follow the script's per-check remediation hint

Gate: C1–C5 all pass.

Test set D — Validate gate

#	Test	How	Pass	Fail action
D1	Evidence report	`UPRTG-Report.ps1`	HTML/CSV report generated with printers, shares, connector, targeting	Re-run; check output path

#	Test	How	Pass	Fail action
D2	Success rate	Review pilot job outcomes	Acceptable success rate, no recurring failures	Root-cause via troubleshooting
D3	Secure release (if used)	QR/badge release test	Job releases correctly	Verify secure-release config
D4	No open Sev-1/Sev-2	Issue log review	None open	Resolve before Expand
D5	Sign-off	Record approver + date	Sign-off captured	—

Gate: D1–D5 all pass; pilot signed off.

Test set E — Expand & retirement gate (per wave)

#	Test	How	Pass	Fail action
E1	Wave targeting	`UPRTG-Set-ScopeTargeting.ps1 -WhatIf` then apply	`UP-WaveN-Users` populated, granted on wave shares	Re-run apply
E2	Wave printers shared	UP → Printers	Each wave printer **Shared**	Re-share
E3	Wave validation	`UPRTG-Validation.ps1` + `UPRTG-Report.ps1`	PASS for wave scope	Troubleshoot before next wave
E4	No residual server dependency	Client print test with old share removed	Prints via cloud, not `\\server\printer`	Remove GPO/mapped print deployment
E5	Retirement safety (parity)	Confirm no reliance on **Delegated Admin** / **macOS** that is not yet in GCC High	Confirmed not blocking	Keep server until GCC High parity confirmed (*verify for GCC High*)

Gate per wave: E1–E5 pass **before** starting the next wave and **before** decommissioning the print server whose printers that wave covered.

Regression / ongoing checks

- **Monthly:** `UPRTG-Report.ps1` for a current inventory; review **Usage and Reports** against the licensed job pool so you don't exhaust it.
- **After any change:** re-run **Discovery** → **Validation** → **Report**.
- **After OS upgrades:** re-check `winver`; devices upgraded **to** Win 11 22H2+ no longer need the registry keys, but leaving them in place is harmless.

Troubleshooting — Universal Print on GCC High

GCC-High-specific symptoms and fixes. Most Universal Print problems on a Government tenant trace back to one of four root causes: **wrong cloud (`.com` instead of `.us`)**, **missing `-Environment USGov`**, **missing US Gov registry keys on older clients**, or **blocked `*.print.azure.us` endpoints**. Work the table for your symptom; each fix references the relevant implementation step.

Re-run `UPRTG-Discovery.ps1` → `UPRTG-Validation.ps1` after any fix to confirm resolution.

T1. Sign-in fails / "account doesn't exist" / redirect error when connecting Graph

- **Likely cause:** Connecting against the **Commercial** identity endpoint. A GCC High account does not exist at `login.microsoftonline.com`.
- **Fix:** Always connect with `Connect-MgGraph -Environment USGov` (all bundled scripts do this). Confirm the sign-in page shows `login.microsoftonline.us`. If you scripted `Connect-MgGraph` yourself, add `-Environment USGov`.
- **Verify:** `UPRTG-Discovery.ps1` completes sign-in and reports UP reachable.

T2. Universal Print blade shows "not available" or licensing banner in the portal

- **Likely cause A — wrong cloud:** You opened `portal.azure.com`. GCC High uses `portal.azure.us`.
- **Likely cause B — licensing:** The admin account has **no eligible UP license**, or licensing hasn't propagated yet.
- **Fix:** Open `portal.azure.us`. In `portal.office365.us` confirm the admin holds a GCC High SKU (`SPE_E3_USGOV_GCCHIGH` / `SPE_E5_USGOV_GCCHIGH` / `Universal_Print_for_GCCHIGH_USGOV_GCCHIGH`). **Wait up to 24 hours** after assigning a license.
- **Reference:** `02-requirements.md` R2/R4; `03-implementation.md` Steps 1, 3.

T3. Printer discovery fails on Windows 10 / older Windows 11 clients (works on Win 11 22H2+)

- **Likely cause:** Pre-22H2 clients are not natively supported in GCC High and are trying

Commercial print endpoints because the **three US Gov registry keys** are missing.

- **Fix:** Deploy to the **user** context (via Intune or user-context script), then have the user

sign out/in:

```
reg add "HKCU\SOFTWARE\Microsoft\Print\MPS" /t REG_SZ /v "OAuthAuthority"    /d "https://login.microsoft
reg add "HKCU\SOFTWARE\Microsoft\Print\MPS" /t REG_SZ /v "MSGraphBaseUrl"    /d "https://graph.microsoft
reg add "HKCU\SOFTWARE\Microsoft\Print\MPS" /t REG_SZ /v "MSGraphResourceId" /d "https://graph.microsoft
```

- **Best fix:** Upgrade the device to **Windows 11 22H2+**, which works natively with no keys.
- **Reference:** `03-implementation.md` Step 8; `gcc-high-parity-notes.md` §4.

T4. Connector won't register or drops offline

- **Likely cause A — endpoints blocked:** Proxy/firewall does not allow the GCC High

`*.print.azure.us` endpoints and `login.microsoftonline.us` / `graph.microsoft.us`.

- **Likely cause B — proxy in user context only:** The connector runs as a **service

(LocalSystem)**; a user-context proxy setting doesn't apply.

- **Likely cause C — cloned host:** A cloned connector host runs in parallel and blocks printing.
- **Fix:** Allow the endpoints in `02-requirements.md` R6. Set the proxy for **LocalSystem** with

`bitsadmin`. If cloned, turn off the clone (or remove the connector from the clone **without** unregistering printers).

- **Reference:** `03-implementation.md` Steps 4–5.

T5. Endpoints unreachable / intermittent print failures

- **Likely cause:** Firewall allows Commercial UP URLs (`*.print.microsoft.com`) but **not** the

GCC High `*.print.azure.us` set.

- **Fix:** Explicitly allow, on both connector host and clients, outbound 443 to:

`print.print.azure.us`, `register.print.azure.us`, `discovery.print.azure.us`, `notification.print.azure.us`, `graph.print.azure.us`, `login.microsoftonline.us`, `graph.microsoft.us`.

- **Verify:** `UPRTG-Discovery.ps1` endpoint check passes.
- **Reference:** `gcc-high-parity-notes.md` §3.

T6. Printer registered but users can't see it

- **Likely cause:** The printer is registered but **not Shared**, or the user's **targeting

group** isn't granted on the share, or the user isn't in the group.

- **Fix:** In `portal.azure.us` → Universal Print → Printers → Share the printer (`UP-Pilot-Printer-01`)

and grant `UP-Pilot-Users`. Confirm the user is a member (`entra.microsoft.us` → Groups). Re-run targeting with `UPRTG-Set-ScopeTargeting.ps1 -WhatIf` then apply.

- **Reference:** `03-implementation.md` Steps 6–7, 9.

T7. Delegated (scoped) admin not working in GCC High

- **Likely cause: Delegated Administration** was **not yet available in GCC High** at

authoring ("coming soon").

- **Fix:** Manage centrally with **Printer Administrator** / **Printer Technician** roles for now.

Verify current GCC High availability before designing a scoped-admin model.

- **Reference:** `gcc-high-parity-notes.md` §6.

T8. macOS users can't print

- **Likely cause: macOS client printing was not yet available in GCC High** at authoring.
- **Fix:** Do not rely on Universal Print for Mac users on GCC High yet, and do not retire a

server that is their only print path. **Verify current GCC High availability.**

- **Reference:** `gcc-high-parity-notes.md` §6.

T9. Print jobs succeed but don't appear in reporting / job pool concerns

- **Likely cause:** Report refresh lag, or misunderstanding of job counting.
- **Fix:** Check `portal.azure.us` → **Universal Print** → **Usage and Reports**. Remember a **single

document = one job** regardless of pages/copies/color/duplex; a job counts only when completed. If the pool is near exhaustion, buy 500/10,000-job add-ons.

- **Reference:** `07-best-practices.md` §2.

T10. "It worked in the lab (Commercial) but not in production (GCC High)"

- **Likely cause:** The lab used Commercial portals/endpoints/Graph; production is GCC High.
- **Fix:** Re-check every URL ends in `.us`, every script uses `-Environment USGov`, endpoints are

`*.print.azure.us`, and licensing is the GCC High SKU. This is the single most common cross-cloud mistake — the **Commercial vs GCC High** table in `01-solution-overview.md` is your checklist.

T11. DoD tenant — nothing works

- **Likely cause: Universal Print is not available in DoD.** This is expected, not a

misconfiguration.

- **Fix:** There is no supported Universal Print deployment in DoD at authoring. Track the

Microsoft 365 roadmap. **Verify for DoD** before assuming a change.

- **Reference:** `gcc-high-parity-notes.md` §1.

Escalation checklist (before opening a Microsoft support case)

1. `UPRTG-Discovery.ps1` output (US Gov Graph reachable? endpoints reachable?).
2. `UPRTG-Report.ps1` current-state report (printers, shares, connector, targeting).
3. Confirm cloud: all portals `.us`, all scripts `-Environment USGov`.
4. Confirm licensing propagated (≥24 h) for the affected users **and** the admin.
5. Client `winver` + registry-key presence for any pre-22H2 device.
6. Endpoint allow-list includes the full `*.print.azure.us` set.
7. Note whether the blocked capability is a known **GCC High parity gap** (Delegated Admin, macOS) rather than a fault.

Best Practices — Universal Print on GCC High

These practices keep a **GCC High** Universal Print deployment correct, compliant, and retirement-ready. They complement the phase gates in `03-implementation.md` and `05-testing.md`.

1. Stay in the sovereign cloud — every time

- **Bookmark only the `.us` portals:** `portal.azure.us`, `portal.office365.us`, `intune.microsoft.us`, `entra.microsoft.us`. Never trust a `.com` tab for GCC High work.
- **Every script and Graph call uses `-Environment USGov`.** Make it a review checklist item — a missing `-Environment USGov` silently targets Commercial and fails for a GCC High account.
- **Allow the `*.print.azure.us` endpoints explicitly** on proxy/firewall, plus `login.microsoftonline.us` and `graph.microsoft.us`. Do not assume Commercial UP URLs cover GCC High — they do not.

2. License discipline (Government-specific)

- **License admins too.** In Government, administrators must hold an eligible UP license to even configure the service — assign it to the operator account first, then wait up to 24 h.
- **Size the job pool from real data.** Each G3/G5 Gov license adds 100 jobs/month to a shared pool; unused jobs expire monthly. Use `UPRTG-Report.ps1` + **Usage and Reports** to watch consumption and buy 500/10,000-job add-ons *before* you exhaust the pool.
- **A job = one document** regardless of pages/copies/color/duplex — factor that into pool sizing.

3. Pilot-first, always gate

• Never publish fleet-wide before the pilot passes. Use `UP-Pilot-Users` as the single pilot gate, then `UP-Wave1-Users`, `UP-Wave2-Users`, … for controlled expansion.

• **Validate each wave** with Discovery → Validation → Report before the next. A green pilot is not permission to skip wave validation.

4. Retire print servers safely

• **Prove before you retire.** Decommission a print server only after every printer it hosted is Shared in Universal Print and validated in a wave.

• **Soak with the server powered but shares removed** (1–2 weeks) so you have an instant fallback before final decommission.

• **Remove the old print path** (GPO print deployment, mapped `\\server\printer`) so clients cannot silently keep using the legacy server.

5. Respect GCC High parity gaps (design around them)

• **Delegated Administration** (scoped Printer Administrator via Entra administrative units) was **coming to GCC High "soon"** at authoring. If you need scoped, per-department admin, **verify current GCC High availability** before designing for it; until then, manage centrally with **Printer Administrator** / **Printer Technician** roles.

• **macOS client printing** was **not yet available in GCC High** at authoring. Do not promise Mac users cloud printing and do not retire a server that is their only path until you **verify for GCC High**.

• **No direct-connect UP-ready printers in GCC High yet** — plan on the **connector** for existing printers.

• Treat every "verify for GCC High" flag as a real pre-deployment task, because GCC High parity changes over time — re-check Microsoft's current Government docs.

6. Connector hygiene

• Use a **dedicated, always-on** host (sleep/hibernate disabled, constant internet, .NET 4.8+).

• **Never clone** a registered connector host — parallel clones interfere and block printing.

• If behind a proxy, set the proxy for the **LocalSystem** account (`bitsadmin`) — the connector runs as a service, not in a user session.

• Name connectors by site (`UP-Connector-Site1`) and size printers-per-connector per Microsoft guidance rather than piling every printer onto one.

7. Client-version strategy

• **Prefer Windows 11 22H2+** for pilot and expansion devices — it works natively in GCC High

with no registry keys.

• Where Windows 10 / older Win 11 must be supported, deploy the **three US Gov registry keys** via **Intune** to the user context; standardize the deployment so no device is missed.

• Devices upgraded to Win 11 22H2+ no longer need the keys; leaving them is harmless.

8. Naming, targeting, and least privilege

• **Create objects once and reuse them** (`UP-Pilot-Users`, `UP-Pilot-Printer-01`, `UP-Connector-Site1`, `UP-<Site>-<Model>-NN`). Consistent names make reports and audits legible.

• Target printers to **groups**, never individuals, and build/maintain those groups repeatably with `UPRTG-Set-ScopeTargeting.ps1` (always `-WhatIf` first).

• Operate day-to-day as **Printer Administrator**; reserve **Global Administrator** for the one-time enable/registration.

9. Change safety

• Every remediation runs `-WhatIf` **first** — review the plan, then apply. `UPRTG-Remediation.ps1` and `UPRTG-Set-ScopeTargeting.ps1` default to `-WhatIf`.

• Re-run **Discovery** → **Validation** → **Report** after any change and keep the dated reports as an audit trail.

10. Monitoring and lifecycle

• Monthly: run `UPRTG-Report.ps1`, reconcile printers/shares/targeting, and check job-pool consumption in **Usage and Reports**.

• Watch Microsoft's Government roadmap/docs for GCC High parity progress (Delegated Admin, macOS, direct-connect printers, DoD availability) and revisit any design that was working around a gap.

Deliverables Summary — Universal Print Deployment & Print-Server Retirement (GCC High)

This solution is a complete, self-guided package for deploying Universal Print on a **GCC High** tenant and retiring legacy print servers. Everything below is included.

Documents

Deliverable	What it gives you
01 — Solution Overview	What the solution does, the Assess→Configure→Pilot→Validate→Expand model, and the full **Commercial vs GCC High — what's different** table (portals, endpoints, licensing, client support) with the availability caveat up front.
02 — Requirements Guide	Ten GCC-High-specific prerequisites (R1–R10), each with **verify** and **implement** steps against `.us` portals, plus a pass/fail checklist.
03 — Implementation Guide	Ordered, click-by-click steps across all five phases using the GCC High portals, with **[GCC High]** caveats inline (registry keys, connector, parity gaps) and a reusable naming convention.
05 — Testing & Validation	Pass/fail test sets A–E, one per phase gate, tied to the bundled scripts.
06 — Troubleshooting	GCC-High-specific symptoms and fixes (wrong-cloud errors, `-Environment USGov`, registry keys, connector, endpoints, licensing propagation).
07 — Best Practices	Ten practice areas: sovereign-cloud discipline, licensing, pilot gating, safe server retirement, parity-gap design, connector hygiene, client strategy, naming, change safety, monitoring.
gcc-high-parity-notes.md	Cited Commercial-vs-GCC-High reference: availability, portals, endpoints, client support, licensing, and confirmed parity gaps — with sources and "verify for GCC High/DoD" flags.
Executive Summary	One-page business-level summary for approvers.
global-requirements-infrastructure.md	Category-level prerequisites shared across all Infrastructure solutions.

PowerShell scripts (all connect to US Gov Graph, `-Environment USGov`)

Script	Type	What it does
UPRTG-Discovery.ps1	Read-only	Confirms UP reachability on GCC High, lists printers/shares/connectors, checks the `*.print.azure.us` endpoints. Run at the Assess gate.

Script	Type	What it does
UPRTG-Validation.ps1	Read-only	Validates the pilot/wave scope against pass/fail criteria (printer shared, targeting granted, connector healthy). Run at Pilot/Expand gates.
UPRTG-Report.ps1	Read-only	Generates a dated HTML/CSV evidence report of printers, shares, connector state, and targeting. Run at the Validate gate and monthly.
UPRTG-Remediation.ps1	Change (default `-WhatIf`)	Safely fixes common configuration drift (e.g., re-grant targeting, re-share). Shows a `-WhatIf` plan first; apply only after review.
UPRTG-Set-ScopeTargeting.ps1	Change (default `-WhatIf`)	Builds/populates pilot and wave targeting groups and grants them on shares, repeatably. Shows a `-WhatIf` plan first.

How the pieces fit the phases

- **Assess:** Requirements Guide + `UPRTG-Discovery.ps1` → Assess gate (Test set A).
- **Configure:** Implementation Steps 3–7 + `UPRTG-Set-ScopeTargeting.ps1` → Configure gate (Test set B).
- **Pilot:** Implementation Steps 8–10 + `UPRTG-Validation.ps1` → Pilot gate (Test set C).
- **Validate:** Implementation Step 11 + `UPRTG-Report.ps1` + `UPRTG-Remediation.ps1` → Validate gate (Test set D).
- **Expand & Retire:** Implementation Steps 12–13 + all scripts per wave → Expand gate (Test set E).

Outcome

Universal Print published on GCC High with the correct sovereign-cloud portals and endpoints, parity gaps handled, pilot-proven, and legacy print servers retired only after their printers are validated in the cloud.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

UPRTG-Discovery.ps1

Purpose: Inventories the current Universal Print footprint on a US sovereign-cloud (GCC High) tenant: printers, printer shares, and connectors as seen by Microsoft Graph US Gov, plus a reachability check of the GCC High Universal Print service endpoints. Read-only: it changes nothing. Use it at the Assess gate to baseline the 'from' state. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\UPRTG-Discovery.ps1 (Graph scopes: Printer.Read.All, PrinterShare.Read.All, PrintConnector.Read.All; -Environment USGov) Optional: -OutputPath 'your-output-folder' to write a CSV inventory. EXPECTED RESULTS Sign-in opens login.microsoftonline.us. The script prints whether Universal Print is reachable, counts of printers/shares/connectors, and PASS/FAIL for each GCC High endpoint. PASS overall = UP reachable and endpoints reachable. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\UPRTG-Discovery.ps1 (Graph scopes: Printer.Read.All, PrinterShare.Read.All, PrintConnector.Read.All; -Environment USGov) Optional: -OutputPath 'your-output-folder' to write a CSV inventory. EXPECTED RESULTS Sign-in opens login.microsoftonline.us. The script prints whether Universal Print is reachable, counts of printers/shares/connectors, and PASS/FAIL for each GCC High endpoint. PASS overall = UP reachable and endpoints reachable. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>

UPRTG-Remediation.ps1

Purpose: Detects and, on approval, fixes common Universal Print configuration drift on a GCC High tenant: a targeting group that exists but is not granted on its share, and empty targeting groups. SAFE BY DEFAULT: runs as -WhatIf; it reports findings and the fix plan and changes nothing until you re-run with -Execute (or -Confirm:$false). It never deletes printers, shares, or groups. HOW TO RUN Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\UPRTG-Remediation.ps1 -ShareName 'UP-Pilot-Printer-01' -GroupName 'UP-Pilot-Users' Apply the reviewed fixes: add -Execute (Graph scopes: PrinterShare.ReadWrite.All, Group.Read.All; -Environment USGov) EXPECTED RESULTS Sign-in opens login.microsoftonline.us. Findings print as OK / DRIFT. In preview, fixes show as 'WhatIf:'. With -Execute, applied fixes print PASS. Re-run UPRTG-Validation.ps1 after. .NOTES White-label — the repeatable way. Changes Microsoft Graph US Gov state only when -Execute is set. #>

How to run: Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\UPRTG-Remediation.ps1 -ShareName 'UP-Pilot-Printer-01' -GroupName 'UP-Pilot-Users' Apply the reviewed fixes: add -Execute (Graph scopes: PrinterShare.ReadWrite.All, Group.Read.All; -Environment USGov) EXPECTED RESULTS Sign-in opens login.microsoftonline.us. Findings print as OK / DRIFT. In preview, fixes show as 'WhatIf:'. With -Execute, applied fixes print PASS. Re-run UPRTG-Validation.ps1 after. .NOTES White-label — the repeatable way. Changes Microsoft Graph US Gov state only when -Execute is set. #>

UPRTG-Report.ps1

Purpose: Produces a dated HTML and CSV report of the Universal Print footprint on a GCC High tenant: printers, printer shares (with granted groups), and connectors. Read-only: it writes report files but changes no tenant state. Use it at the Validate gate for sign-off evidence and monthly for lifecycle review. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\UPRTG-Report.ps1 (Graph scopes: Printer.Read.All, PrinterShare.Read.All, PrintConnector.Read.All; -Environment USGov) Optional: -OutputPath 'your-report-folder' (defaults to the current folder). EXPECTED RESULTS Sign-in opens login.microsoftonline.us. The script writes UPRTG-Report-<timestamp>.html and matching CSVs, and prints the output paths. Open the HTML for a readable snapshot. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\UPRTG-Report.ps1 (Graph scopes: Printer.Read.All, PrinterShare.Read.All, PrintConnector.Read.All; -Environment USGov) Optional: -OutputPath 'your-report-folder' (defaults to the current folder). EXPECTED RESULTS Sign-in opens login.microsoftonline.us. The script writes UPRTG-Report-<timestamp>.html and matching CSVs, and prints the output paths. Open the HTML for a readable snapshot. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>

UPRTG-Set-ScopeTargeting.ps1

Purpose: Creates (if missing) a Microsoft Entra security group used to target a Universal Print share, adds the specified members, and grants that group access on the named share. This is the repeatable way to build pilot and wave targeting on a GCC High tenant. SAFE BY DEFAULT: runs as -WhatIf; it prints the plan and changes nothing until you re-run with -Execute (or -Confirm:$false). HOW TO RUN Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\UPRTG-Set-ScopeTargeting.ps1 -GroupName 'UP-Pilot-Users' -ShareName 'UP-Pilot-Printer-01' -MemberUpn 'user1@your-tenant.us','user2@your-tenant.us' Apply: add -Execute (Graph scopes: Group.ReadWrite.All, PrinterShare.ReadWrite.All, User.Read.All; -Environment USGov) EXPECTED RESULTS In preview you see each action prefixed 'WhatIf:'. With -Execute the group is created/updated, members added, and the group granted on the share; the script prints PASS per action. .NOTES White-label — the repeatable way. Changes Microsoft Graph US Gov state only when -Execute is set. #>

How to run: Preview (default, no changes): powershell -ExecutionPolicy Bypass -File .\UPRTG-Set-ScopeTargeting.ps1 -GroupName 'UP-Pilot-Users' -ShareName 'UP-Pilot-Printer-01' -MemberUpn 'user1@your-tenant.us','user2@your-tenant.us' Apply: add -Execute (Graph scopes: Group.ReadWrite.All, PrinterShare.ReadWrite.All, User.Read.All; -Environment USGov) EXPECTED RESULTS In preview you see each action prefixed 'WhatIf:'. With -Execute the group is created/updated, members added, and the group granted on the share; the script prints PASS per action. .NOTES White-label — the repeatable way. Changes Microsoft Graph US Gov state only when -Execute is set. #>

UPRTG-Validation.ps1

Purpose: Validates that a Universal Print deployment scope on a GCC High tenant is correctly configured before you advance a phase gate: printer registered and shared, targeting group present and granted, and connector healthy. Read-only: it asserts state, it does not change it. Run it at the Pilot and Expand gates. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\UPRTG-Validation.ps1 (Graph scopes: Printer.Read.All, PrinterShare.Read.All, PrintConnector.Read.All, Group.Read.All; -Environment USGov) Optional: -ShareName 'UP-Pilot-Printer-01' -GroupName 'UP-Pilot-Users' to scope checks. EXPECTED RESULTS Sign-in opens login.microsoftonline.us. Each check prints PASS or FAIL with a hint. OVERALL PASS = the scope is ready to advance past its gate. .NOTES White-label — the repeatable way.

Read-only against Microsoft Graph US Gov endpoints. #>

How to run: powershell -ExecutionPolicy Bypass -File .\UPRTG-Validation.ps1 (Graph scopes: Printer.Read.All, PrinterShare.Read.All, PrintConnector.Read.All, Group.Read.All; -Environment USGov) Optional: -ShareName 'UP-Pilot-Printer-01' -GroupName 'UP-Pilot-Users' to scope checks. EXPECTED RESULTS Sign-in opens login.microsoftonline.us. Each check prints PASS or FAIL with a hint. OVERALL PASS = the scope is ready to advance past its gate. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph US Gov endpoints. #>